

AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL

Submission: model_8

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

The solution proves that the a_i share a common residue modulo g , meeting the partial-progress milestone. The upper bound is not established, however: it never rules out $k \geq 4$, and in the $k = 3$ analysis it gives no proof bounding the cofactor cases $m \geq 3$ (only $m = 1, 2$ are treated, with a single example for $m = 5$) — both instances of **NEC3**. There are also lesser omissions: the common residue is asserted to be nonzero without proof (**ME2**), and the constructed example's gcd is not shown to equal 998285 (**ME4**). It therefore scores **1** for the common-residue milestone.